Birth Control on Every Corner: The Impacts of Pharmacists' Prescription of Birth Control

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June 30, 2026

Abstract

In 2016, Oregon became the first state to allow pharmacist prescription of hormonal contraceptive pills. I find that births in treated zip codes decrease by nearly 9% per year following legalization of pharmacists' prescription of birth control. The effect is somewhat stronger when focusing on zip codes with a history of Medicaid physician shortages. Quarterly, state-level Medicaid prescriptions data show an increase in birth control prescriptions that is persistent after a delayed spike. Using county-level monthly data, I show the reduction in births is accompanied by an initial reduction in abortions in the first year and a continued decline in births only when focusing on provider-shortage counties. This evidence suggests that pharmacists can play an important role in alleviating problems caused by health worker shortages, and that analyzing localized effects is important in the reproductive health context.

JEL Classification: I18, J13, J18

Keywords: abortion, childbearing, sexual health, access to contraceptions

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Thank you to participants at Southern Economic Association meetings, Tamara McGavock, Logan Lee, and Bradley Hall for useful comments and support.

Data come from publicly available sources and replication files are available on request. The author has no conflicts of interest nor funding to report for this project.

1 Introduction

Access to contraception and other reproductive control technologies is constantly shifting in the modern legal landscape. Beginning in the mid-2010s, localized efforts to reduce barriers to hormonal contraception were seen in many parts of the country. A reduction in barriers to methods like the birth control pill included the emergence of telehealth and shipping discreet packages to consumers as well as the push for non-physician prescriptions of hormonal contraception. Home delivery of the pill through online services — and telehealth services more generally — rose in popularity during the COVID-19 pandemic. In 2024, over-the-counter birth control pills entered the market. Well before these changes, though, some states were working to reduce patients' need to physically visit a physician's office for birth control and began to allow pharmacists to prescribe some of the most common forms of birth control: oral contraception ("the pill") and the monthly hormonal contraceptive injection ("the shot").The first locality to move toward this expansion in access was Oregon, which passed a law permitting pharmacists' prescription of hormonal contraception to take effect in January of 2016.[1]

Economics literature has long documented the impact of access to reproductive control technology — such as birth control pills, long-acting reversible contraception such as IUDs or implants, or abortion — but much of the work focuses on the monumental change from extremely limited access to somewhat easier prescription access. For example, Goldin and Katz (2002) demonstrate that women born in times and locations in which the birth control pill was accessible to young, single women were more likely to obtain higher education and marry later than their peers born in slightly earlier cohorts or in states with stricter regulations around the pill. Other studies, namely Ananat and Hungerman (2012), Bailey (2006), and Bailey (2012), add to our understanding from Goldin and Katz (2002) that access to the pill had a major impact on women's and infants' outcomes, and that many of those effects persist into reductions on poverty, next-generation health outcomes, and labor market outcomes. These studies document the impact of new access

[1]Pharmacists' prescription authority for the shot began in Oregon in 2017. In 2016, Oregon also required all insurance to cover up to a year of contraceptive care — a move that reduced the frequency a patient needed to refill their contraceptive prescriptions.

to the pill for young, single women in the 1970s — but does an increase in access to family planning technology in 2016 have similar impacts to that in the mid- to late-twentieth century?

I find that the ability of pharmacists to prescribe birth control pills in Oregon reduced unintended pregnancies — through large, consistent effects on annual birth counts at the zip code level and through a reduction in abortions in the first year of pharmacist prescription. I identify that these effects are driven by areas experiencing Medicaid primary care shortages, suggesting that this sort of scope-of-practice expansion is especially helpful in areas struggling to supply an adequate health care labor force. I also show that county-level birth estimates are unable to demonstrate the same results as the zip code-level estimates, which supports the idea that county-level data inadequately identify the treated and control group geographies.

Although this policy is specifically related to reproductive health care, it also speaks to a broader question regarding provision of health care in spaces with limited provider supply. Demographic changes — including the aging baby boomer population and consistently falling fertility rates — have contributed to stark health care worker shortages across the United States. The issue of provider shortages is more prominent in rural areas than in urban ones — with over 90% of rural counties (compared to 74.4% of non-rural counties) having a shortage of healthcare workers in 2023 jec (2024). Many states have responded to such provider shortages by expanding the practice and/or prescribing authority of other qualified medical professionals, like registered nurses (RNs), nurse practitioners (NPs), physicians assistants (PAs), and pharmacists.

The literature around these expansions of authority tend to circle around the same results: patients are able to access more care and health outcomes are improved. McMichael (2025) studies expansion in provider authority for NPs and documents a reduction in hospitalizations that could have been avoided with timely outpatient care, likely driven by 'healthier' cases being treated outpatient before arriving at a hospitalization. Similarly, Bae and Sigaud (2026) examine the impact of scope-of-practice expansions to optometrists to do some forms of laser surgery, finding an increase in services provided — driven by those provided by optometrists rather than ophthalmologists — and document a reduction in eye care deserts and wait times for patients. More directly connected to this study, two other works document the impact of increased prescription power by pharmacists. First, Hoagland and Wang (2025) find that pharmacists' ability to prescribe care for minor ailments increased visits to both pharmacies and other providers, and that the effects were concentrated among poorer neighborhoods. Second, Shakya et al. (2025) study a change in prescribing power in Idaho that allowed pharmacists to prescribe select medications, including rescue inhalers and insulin pens. They show that this switch increased Medicare Part D prescriptions and improved timely access to time-sensitive care. In each of these cases, expanding providing or prescribing scope improves patient access — often among those who need it most. Finally, Grossman et al. (2025) study state-level pharmacist prescribing laws and identify a reduction in births in states with pharmacist prescription of birth control. I contribute to this conversation by identifying that birth effects come from directly-treated zip codes, not entire counties, and from areas experiencing provider shortages.

The rest of this paper is organized as follows: I first introduce the background of the Oregon House Bill that established the legality of pharmacist prescription of birth control and discuss other states' implementation of similar policies and existing evidence related to them. Next, I will describe my data and identification strategy, followed by my main results, heterogeneity tests, and a comparison to county-level analyses. Finally, I will conclude and discuss the relevance of this sort of policy change in the production of both reproductive and non-reproductive health care.

2 Background

In June of 2015, Oregon Governor Kate Brown signed into law Oregon House Bill (HB) 2879, which streamlined a process to allow pharmacists to prescribe birth control. Through this initiative, licensed pharmacists could complete a board-approved training program to begin prescribing hormonal contraceptive patches and oral contraception starting in January of 2016.[2] This made Oregon the first-mover among the United States in increasing access to hormonal contraception through pharmacists. California did so later in 2016, and as of 2026, more than 30 states have moved to allow pharmacists to prescribe hormonal birth control pills, with some variation in the stipulations around this prescriptive behavior.

Pharmacists in Oregon seemed to be excited to participate — before the bill even took effect on January 1, 2016, nearly 150 pharmacists had completed their board-certified requirements to prescribe birth control, and within the first year, over 800 pharmacists had completed their training (Rodriguez et al., 2016). As of December 2018, over 1,300 of the state's pharmacists[3] had passed the requirements to prescribe birth control. Additionally, in 2017, the state passed another house bill, HB 2527, allowing pharmacists to provide injectible hormonal contraception ("the shot"), further expanding access to contraception. Most pharmacists who had already begun prescribing birth control completed the requirements to administer the shot, and in 2017 the Board of Pharmacy in Oregon rolled out a training unit that combined the training requirements for the pill and the shot (Jones, 2023).

In state-level work analyzing the first 13 states to adopt such policies compared to the 15 to adopt similar policies in the 2020s, Grossman, Ray, and Wadsworth find pharmacist prescription of contraception to reduce quarterly births by 0.5 births per 1,000 women aged 15-49, with effects driven by those aged 25-34 and 40-44 and those with high school or less education (2025). My results build on this work: I use zip code level data and find that pharmacist prescription of birth control reduced births in treated zip codes by an average of 9% per year, or approximately 3 fewer births per 1,000 women each year for each treated zip code. The number of zip codes per county ranges from 1-38; the

[2] In Oregon, a working group representing a variety of stakeholders met to create a standard procedure for the state's pharmacists to prescribe hormonal contraceptives while maintaining adherence to existing health guidance. The group decided on a five-hour online training module that "covers general information on contraception, medically necessary screenings, and referrals" and established a process for referrals for patients who had medical contraindications or could not afford the pharmacists' fees (Jones (2023); Rodriguez et al. (2016)).

[3] Current administrative data of prescribing pharmacists are not available to researchers, but a 2016 publication noted that of the 3,041 pharmacists practicing in Oregon, only 1,579 worked in retail pharmacies, which would be the individuals most likely to increase access for patients. Of those 1,579, at the time of publication, over 1,200 of those had registered for the licensing course (Rodriguez et al., 2016).

share of treated zip codes within a county ranges from 0-84%, with an average share of treated zip codes of 18.9%.[4] Despite looking at different populations, this estimate is quite similar to that found in (Grossman et al., 2025): if those 3 annual births per 1,000 women were evenly spread across the year, they would represent .75 fewer quarterly births per 1,000 women (which falls within (Grossman et al., 2025)'s estimate's 95% confidence interval). The very slightly larger estimate found here may be a result of focusing on a smaller geographic level (for whom the pharmacists' prescribing ability may matter most) or perhaps a result of focusing on areas experiencing physician shortages. In both study settings, expanding the scope of practice for pharmacists does seem to reduce unplanned pregnancies.

3 Empirical Approach

3.1 Data

To measure birth outcomes, I use publicly available data from the Oregon Health Authority which capture annual, zip code-level birth counts within each county from 2006-2024.[5] To test for effects on abortions, I rely on monthly county-level abortion data — aggregated to the county-quarter — for all 36 counties in Oregon from 2014-2019.[6][7] I also use monthly county-level birth counts aggregated to the county quarter to identify whether a county-level analysis would be able to detect any effects.

I would like to highlight the differences between defining treatment at the zip code level and at the county level to highlight the importance of focusing on zip code data. Zip codes are a slightly odd geographic indicator — they do not adhere to state or county lines, as they are generated by the US Postal Service for the purpose of organizing mail routes. Zip codes are smaller than counties (i.e. each county has multiple zip codes[8]) and therefore

[4]Because some zip codes cross county lines, and the zip code birth data are observed at the zip code by county level, and because many some zip codes contain zero women aged 18-44 (the closest approximation available to "childbearing age" at the zip code level), the number of observations left in the analyses is not equal to the total number of zip codes in Oregon over the study period.

[5]Data are available back to 1989.

[6]I use this range because Oregon's Medicaid expansion began in 2014 and COVID-19 arrived in 2020.

[7]I present quarterly results for county-level estimates to reduce the prevalence of zeros.

[8]Some zip codes cross county lines, resulting in multiple zip code by year observations (but only one

provide a more precise location than counties. Because the change in access resulting from a pharmacist prescribing birth control is unlikely to have a vast geographic reach, I expect the zip code birth analyses to better reflect the true treatment effects. Figure 1 shows the zip codes and counties in Oregon. Zip codes with a documented pharmacist prescribing birth control are shown in dark/mossy green, and counties identified as 'treated', based on having at least one zip code with a prescribing pharmacist, are shaded in green. Zip codes without a prescribing pharmacist are outlined and diagonally striped. Table A1 shows the issue in another format: if we use the county treatment assignment, we see 764 zip code by county observations without licensed providers are being considered 'treated'; only 37 such units are considered 'comparison' zips. We can see from these exhibits that many untreated zip codes are being lumped into the treated group due to the county definition of treatment. Because the effects are likely to be very localized, the annual birth counts at the zip code level will be the primary focus of this study. The county-level analyses contribute to our understanding by including more granular times (months/quarters rather than years), including abortion counts, and documenting how differently estimates would appear if we used a more aggregated geographic indicator to establish treatment and comparison groups.

To identify treated and comparison geographies, I use a map of participating pharmacies available online (Birth Control Pharmacies, 2018).[9] I use the earliest available internet archive of this map to identify which zip codes had a pharmacy that publicly reported offering pharmacist prescription of birth control. This public-facing and easily accessible website helpfully identifies information that potential patients may have had available to them at the time of contraceptive decision-making.[10]

To identify areas that might most benefit from pharmacist prescription for birth control, I use Medicaid Primary Care Workforce data from 2016 from the Fitzhugh Mullan

zip code by county by year observation).

[9]See Figure A1 for a screenshot of the website.

[10]In addition to the publicly available map data, I use an administrative record of pharmacists' licensing dates through 2018. These data identify individual pharmacists, the address of their employer, and the date they completed the necessary training. However, an updated version of this data is not available and the data are no longer discoverable online or through data requests. This dataset does align with the data available from the map online and shows that most ever-treated zip codes became treated in 2016.

Institute for Health Workforce Equity. These data identify the provider to patient ratio, identifying which primary care providers take Medicaid patients and the county's Medicaid population. Using these data, I identify a county as a "Primary Care Shortage" area if the patient-provider ratio is in the highest ranking group. These counties, nationwide, have a patient:provider ratio of 603-3900:1.[11]

Finally, I use quarterly state-level Medicaid claims data to measure whether birth control prescriptions among Medicaid-covered patients change as a result of pharmacist prescribing. Figure A2 shows Medicaid-reimbursed units for all categories of contraceptive products and shows an increase in contraceptive care in late 2016 — with an increase in contraceptive pills and IUDs. The level of pill prescriptions reimbursed by Medicaid does not change drastically at the start of pharmacist prescriptions in 2016, perhaps suggesting that any changes in pill prescription initially came from privately insured patients or took time to appear in claims data. It is also worth noting that the spike in Quarter 2 of 2017 may be a result of increased concern around reproductive health care access after the surprising-to-many results of the 2016 election — a phenomenon documented through analysis of Google trends and early commercial claims data (Nobles et al. (2019), Pace et al. (2019)).

3.2 Identification Strategy

I use a Poisson model to estimate effects on birth and abortion counts, following Lindo et al. (2019). This model is more appropriate than a linear model because zip code-year (and county-month) observations are generally small counts and have somewhat frequent instances of zeros. I use a difference-in-differences approach to estimate the causal effects of pharmacist prescription of birth control. This approach exploits within-geography variation over time and controls for aggregate time shocks, as well as fixed differences across geographies over time and differences in pre-regulation trends. In order for this approach to be valid, it must be true that changes in births (and abortions) for com-

[11] These data are separated into 5 'ranks' based on how the county compares to national data. Oregon's county with the highest patient:provider ratio in 2016 was Multnomah County, with a patient:provider ratio of 746:1.

parison zip codes (or counties) provide a good counterfactual for the changes in births (or abortions) that would have been observed for treated geographies, if access to birth control remained unchanged – or, rather, if the pharmacists had not become eligible to prescribe birth control and taken the appropriate licensure. My approach to estimating the effects of changes in availability of pharmacist prescription of birth control on abortions and births corresponds to the following equation:

$$E[y_{gt}|Pharm_{gt}, \alpha_g, \alpha_t] = exp(\theta_1 Pharm_{gt} + \alpha_g + \alpha_t) \tag{1}$$

where y_{gt} is the outcome of interest for residents of geography g in time t; $Pharm_{gt}$ is an indicator for whether geography g had a pharmacist licensed to administer birth control in time t; α_g are geography fixed effects; α_t are time fixed effects.[12],[13] I use 2016 zip code population data as the Poisson's exposure variable and all reported standard-error estimates are clustered on the geography to account for correlation within geographies over time.

While it might appear that a dynamic framework using staggered difference-in-differences tools may be appropriate, I argue that the static approach is more fitting for two reasons: first, the vast majority of zip codes had their first pharmacist licensed in 2016 — 84% of the total treated zip codes become treated in 2016, and the share rises further to 88% when I exclude zip codes that do not experience Medicaid physician shortages. Because zip code birth data are annual counts, any more granular variation (i.e. start dates varying within 2016) cannot be used in the analysis. Second, county-level data, which are monthly counts, absorbs much of the variation in treatment: in the county-level comparison, only 5 of 36 counties (or 13.8%) never experience pharmacist prescription licensure. When using zip code level data, we instead see approximately 20% of zip codes remain untreated. When restricting to physician shortage areas, the problem becomes

[12] I use 2016 zip code population data from the United States Census Bureau to identify zip code population that are female aged 18-44. Due to the limited zip code level data available, this is my best approximation at zip code female population of childbearing age.

[13] County time-variant controls, which may be added to the above model as X_{gt}, include the share of the population that are Black, Hispanic, aged 15-44, female, and unemployed. Geographic controls are only consistently available at the county-level and will not be included in the zip code birth analyses.

even clearer: no comparison counties are classified as experiencing physician shortages for the Medicaid population, while roughly 16.5% of zip codes that experienced shortages are zip codes lacking a licensed prescribing pharmacist.

4 Results

I identify a zip code as 'treated' if it has a prescribing pharmacist operating in their zip code; all other zip codes are in the comparison group. Panel A of Figure 2 shows that pharmacist prescription of birth control reduced births, and that effects begin to show up in the first year and grow in magnitude and significance throughout the remainder of the study period. The point estimates for the average treatment effect are shown in Table 2. The comparison using all existing zip codes does produce statistically significant pre-period outcomes (though mostly flat) pre-period estimates, suggesting the full-sample comparison may not demonstrate parallel trends. When we focus, instead, on zip codes that exist in Medicaid physician shortage areas, the pre-2016 estimates are no longer statistically different from zero. We still see a visually compelling decrease in annual births in 2016 and beyond, with an average treatment effect of roughly 9%. The estimated effects in the restricted sample are noisy — which is unsurprising given the dramatic reduction in sample size — but the approximately 9% reduction in annual births is statistically significant at the 10% confidence level. Prior to treatment, zip codes in the treated group experienced 200 annual births per zip code; pharmacist prescription of birth control may have reduced 20 births per year per zip code.

4.1 Heterogeneity

While the main results indicate a reduction in births at the zip code level resulting from pharmacist prescription of birth control, I also want to test for effects among those more likely to be impacted by this change in access. To do so, I test for effects based on the county's availability of Medicaid primary care providers. Using Fitzhugh Mullan Institute rankings for counties based on their patient:provider ratio, I created "shortage" indicators for all counties with patient:provider ratios greater than 603:1. I then drop any treated counties that do not have a provider shortage, effectively comparing zip codes with a Medicaid provider shortage *and* a pharmacist prescriber to zip codes with no pharmacist prescribers.[14] The results from this test are presented in Panel B of Figure 2 and the average treatment effect estimates can be found in Column 2 of Table 2. Restricting the treated group to only those zip codes in counties with provider shortages demonstrates a stronger result: pharmacist prescription of birth control reduced annual zip code births by as much as 17.9%. These results are immediately statistically significant, unlike in the main results which were not statistically significant in the first three years. They also grow more quickly than in the main results, supporting the idea that people living in communities with primary care shortages may have 'more to gain' from pharmacist prescription than those living in areas with easy access to providers.

4.2 County-Level Estimates

Finally, I test the robustness of my earlier findings to an estimation of county-level data.[15] Results for both births and abortions can be seen graphically in Figure 3 and the corresponding point estimates are in Table 3. The top panels estimate effects on births; the bottom panels estimate effects on abortions. The left-hand panels in Figure 3 include demographic controls (shares of the county population that are Black, Hispanic, aged 15-44, female, or unemployed), while the right-hand panels include those controls but consider counties as treated only if they have a prescriber *and* experienced a Medicaid physician shortage.[16] Columns 1 and 5 present estimates with no demographic controls, Columns 2 and 6 control for county-level unemployment rate and population shares. Columns 3 and 7 drop any counties from the treated group that did *not* experience a physician shortage in 2016. Because Figure 3 demonstrates different effects on abortion in

[14] The results shown are using the strictest definition of physician shortage and eliminate 26 otherwise-treated counties (and their corresponding zip codes from the study. Results using more lenient definitions of physician shortage follow similar patterns and are available in the Appendix Figure A3.

[15] County-level data exist at the monthly level, and results shown here are analyzed at the quarter level to reduce the number of observations with a zero count.

[16] Note that this is different from the zip code shortage analyses: if we only compare geographies in shortage areas to one another, we have no comparison counties. Instead, this comparison drops non-shortage, treated counties.

different post-periods, I also estimate the effects on both births and abortions separating the post-period into the first four quarters and the following quarters. These point estimates can be found in Columns 4 and 8 of Table 3.

The results suggest that pharmacist prescription of birth control had no meaningful, statistically significant, detectable effect on births at the county level in the first 4 years — outside of an increase in the first quarter of 2017 and a smaller decrease in the third quarter of 2019.[17] On the other hand, immediately after pharmacist prescription of birth control comes into play, abortions fall, though the effect fades after one year. The estimated effect on abortion reduction in the first year varies with specification choice — likely as a result of the limited number of counties left in the sample when considering different thresholds for primary care shortages. First-quarter effects on abortion are plausible: if a person were to begin taking the pill correctly in January or February, they may avoid an unintended pregnancy that would have ended in abortion by the end of March, as the vast majority of abortions occur *within* the first 6-9 weeks of gestation.[18] The estimates are noisy — likely a result of wide monthly variation in abortions per county — but appear to be negative in the first year. Whether looking at the primary care shortage sample or the full sample, the pre-treatment estimates are often not statistically significant, though they follow odd patterns and may suggest some differences in pre-period trends when analyzing the county-level data.

Taken with the results discussed in the previous section, pharmacist prescription of birth control had a meaningful impact when looking at births at the zip code level and the first year of abortions at the county level. This suggests that pharmacist prescription of birth control can be a helpful tool for people to manage their fertility, but that the geographic reach of a prescribing pharmacist may be relatively small.

[17] The first quarter is also statistically significant and positive, but birth effects within three months of pharmacist prescription seem spurious.

[18] According to the Oregon Health Authority's abortion reports, nearly 70% of all abortions occurring in Oregon between 2014-2016 occurred during the first 9 weeks of gestation. According to the CDC's Abortion Surveillance report in 2016, 47.7% of all abortions in Oregon occurred in the first 6 weeks of gestation.

5 Discussion

Taking the results of this draft, it is apparent that pharmacists' ability to prescribe birth control in Oregon reduced unintended pregnancies in Oregon. At the zip code level, births fell immediately and continued to fall in the post-prescription world relative to what would have been expected without a prescribing pharmacist in the zip code. This policy also reduced abortions in counties with prescribing pharmacists within the first year pharmacists began prescribing. Births at the county-quarter level did not demonstrate the same large and persistent reduction as in the zip code analyses, suggesting that identifying treated groups based on county may be too broad and that effects may be quite localized to the prescribing pharmacist. When we narrow the comparison to zip codes that experienced Medicaid primary care shortages in 2016, the effects are even stronger and are robust to varying definitions of 'shortage'.

The results together show that increasing prescribing power of pharmacists (or perhaps other non-physician health care providers) could be an effective strategy in combating health care provider shortages. Of all US counties, approximately 95% are experiencing a shortage in primary care physicians; this number jumps to 97% of non-metropolitan counties (RHI, 2023). As the baby boomer generation ages out of the labor force, states are grappling with challenges related to demand for health care exceeding supply, with the gap between demand and supply projected to increase. Shifting some of a primary care physician's labor to another qualified health care professional could relieve some of that shortage. Allowing pharmacists to prescribe things like hormonal contraception is one such approach — expanding prescriptive or general providing privileges to other professionals, like physicians' assistants (PAs) or registered nurses (RNs) is another. Using data from 2017, estimates suggest that approximately 180 physician hours were diverted to pharmacists every two years, suggesting a potential reduction in over 700 hours of physician labor (Jones, 2023).[19] This suggests that pharmacists' prescription of contra-

[19] Data suggest that 10% of all Medicaid users who started on the pill between 1/1/2016 and 12/31/2017 did so using a pharmacist (367 individuals in 2016 and 2017). If all of those patients would have gone to a physician instead, and birth control appointments typically take 30 minutes, and new patient take-up remained consistent across all 9 years since the start date, 810 physician hours or more were diverted to pharmacists.

ception (and potentially other "low-risk" medications) could be one low-cost avenue to address concerns around demand for physician care exceeding supply.

It is worth noting that even if an area has pharmacists who are able to prescribe and that area has no detectable reduction in abortion or birth rates, it is entirely possible that increased access to birth control still creates benefits for area residents. Increased access to the pill through pharmacists could improve women's non-reproductive outcomes — patients may spend less time searching for, waiting for appointments with, and seeing medical professionals; patients may spend less on transportation or time off of work; patients may receive psychological benefits from working with a pharmacist rather than a physician to address their reproductive health concerns. None of these benefits would show up in the birth or abortion data analyzed here. Additionally, failure to detect a reduction in births or abortions — and, in fact, detecting an increase in births — *could* be evidence of increased reproductive freedom.

Some health policy implementation may be 'small' in terms of identifiable magnitude in population-level data — detecting noisy or zero effects at higher geographic levels than the treatment occurs should therefore be interpreted with caution. Additionally, 'small' interventions may have significant and meaningful impacts that cannot be detected by the data commonly collected and analyzed in our field. Measuring convenience, comfort, or time saved could provide further evidence of improvement in settings with more incremental policy changes. Future studies undertaking the task of estimating effects from very localized policy implementations should be cautious in describing null results — especially if granular data are not available.

Pharmacist prescription of birth control has potential to mitigate the damage of restrictions on reproductive care (such as the overturning of Roe v. Wade or decreased funding to Title X clinics), ease health care shortages, improve patients' non-reproductive outcomes, and increase reproductive autonomy. Oregon's venture into pharmacist-prescribed birth control provides evidence that a shift in provision of care like this can improve the directly-impacted outcomes — especially among populations that otherwise lack sufficient care provision.

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Tables

Table 1
Summary Statistics by Treatment Timing Year

	Full Sample			
	Never Treated	2016	2017	2018
Births (2007-2015)	21.11	190.44	59.40	29.90
	(39.22)	(217.85)	(54.34)	(35.96)
Births (2016-2024)	27.10	171.40	53.14	29.68
	(66.08)	(176.67)	(45.43)	(34.42)
Female Population 18-44, in 2016	485.40	5292.48	765.00	459.57
	(1091.18)	(3252.46)	(540.80)	(499.28)
Number of Zips	247	210	8	7

	Restricting to Physician Shortage Areas			
	Never Treated	2016	2017	2018
Births (2007-2015)	16.73	195.70	29.78	
	(34.01)	(249.75)	(7.93)	
Births (2016-2024)	36.74	159.69	35.11	
	(108.51)	(189.47)	(4.48)	
Female Population 18-44, in 2016	1039.44	6096.00	537.00	
	(2058.74)	(3323.51)	(0.00)	
Number of Zips	28	88	1	0

Notes: This table summarizes births prior to and after pharmacists were allowed to prescribe birth control by treatment timing group. It also demonstrates the female population aged 18-44 in that zip code in 2016 and the number of zip codes in the group. The top panel includes all zip code by county observations, and the bottom panel includes only zip codes in a Medicaid physician shortage area.

Table 2
Estimated Effects of Pharmacist Prescription on Births – Zip Code Annual Birth Data

	(1)	(2)	(3)	(4)	(5)	(6)
Average Treatment Effect	-0.112***	-0.105***	-0.111***	-0.105*	-0.101*	-0.093*
	(0.026)	(0.023)	(0.026)	(0.057)	(0.055)	(0.053)
Pre-Treatment Average	199.88	199.88	200.73	202.15	202.15	202.15
Observations	8496	8496	8226	2106	2106	2088
Physician Shortage Areas	N	N	N	Y	Y	Y
County Linear Time Trends	N	Y	N	N	Y	N
Excluding 'Late Start' Treated	N	N	Y	N	N	Y

Notes: Results are from a Poisson model including zip code and year fixed effects. "Physician Shortage Areas" indicates the analyses use only zip codes in Medicaid physician shortage areas.
*, **, and *** indicate statistical significance at the 10, 5, and 1% levels, respectively.

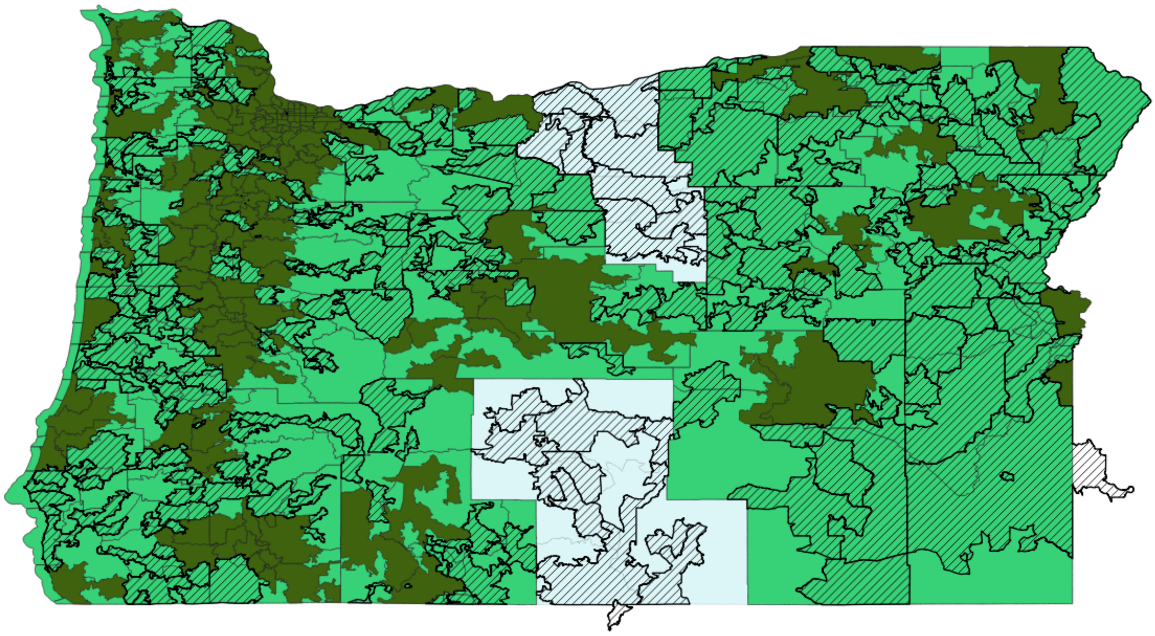
Table 3
Estimated Effects of Pharmacist Prescription on County-Level Births and Abortions

	Births				Abortions			
Average Treatment Effect	0.029	0.022	0.042		0.020	-0.123	-0.604***	
	(0.060)	(0.077)	(0.098)		(0.045)	(0.078)	(0.112)	
Average Treatment Effect in 2016				0.071				-0.827***
				(0.093)				(0.112)
Average Treatment Effect 2017-2019				0.017				-0.269
				(0.106)				(0.305)
Observations	864	864	240	240	864	864	240	240
Population Controls	N	Y	Y	Y	N	Y	Y	Y
Physician Shortage Areas	N	N	Y	Y	N	N	Y	Y

Notes: Results are from a Poisson model including county-level controls and quarter, year, and county fixed effects. Population controls include share of the county population that is Black, Hispanic, 15-44 years old, female, and unemployed. "Physician Shortage Areas" indicates the analyses use only counties with providers *and* physician shortages as treated; all other counties are in the comparison group. *, **, and *** indicate statistical significance at the 10, 5, and 1% levels, respectively.

Figures

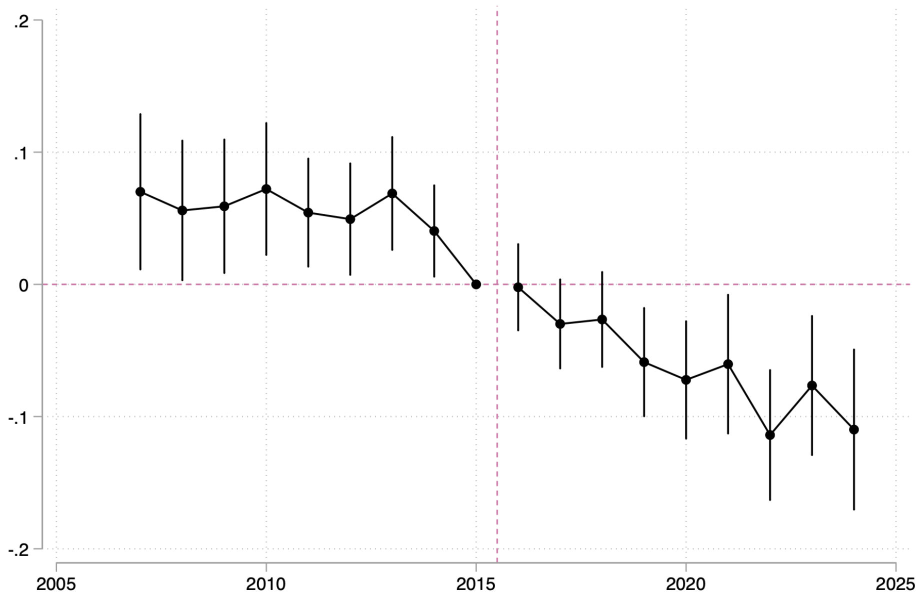
Figure 1
Location Map – Zips and Counties



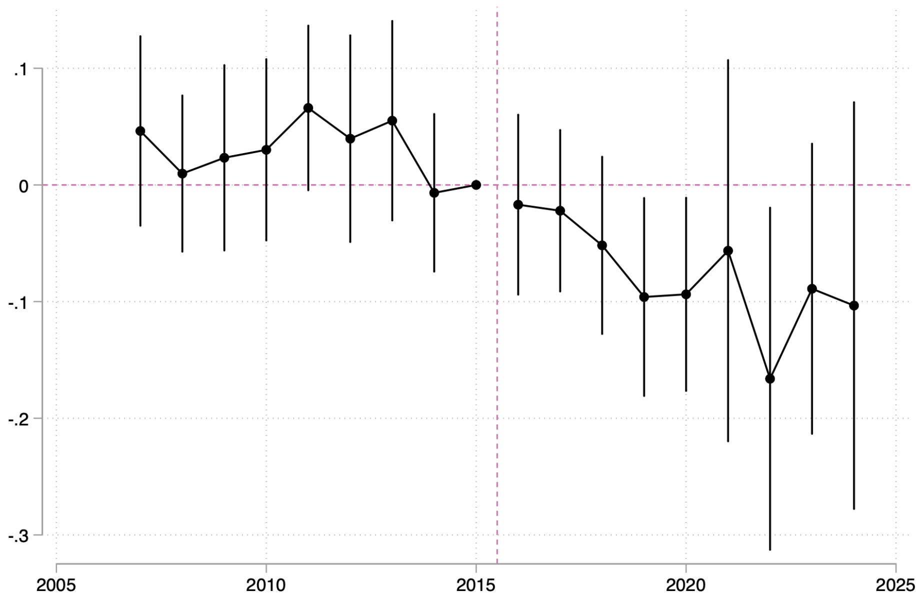
Notes: This map indicates the counties that are treated in bright green, with an overlay of the treated zip codes (based on administrative and website data) in a moss green on top. The diagonally-striped zip codes are those that do *not* have a listed pharmacy or pharmacist prescribing birth control; zip codes that are shaded green and are diagonally-striped are zip codes that are not treated but are being included in the treatment definition at the county level.

Figure 2
Estimated Effects on Births – Annual Zip Code Birth Data

(a) Births



(b) Births, Medicaid Physician Shortage

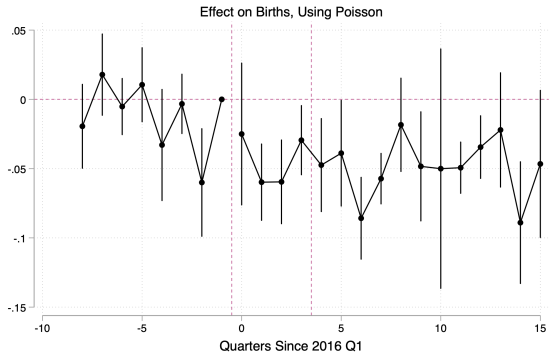


Notes: These figures plot the estimated effects of pharmacist prescription of birth control on births and abortions using a Poisson model. The top panel uses the full sample and the bottom panel restricts to areas with Medicaid physician shortages. The models control for zip code and year fixed effects.

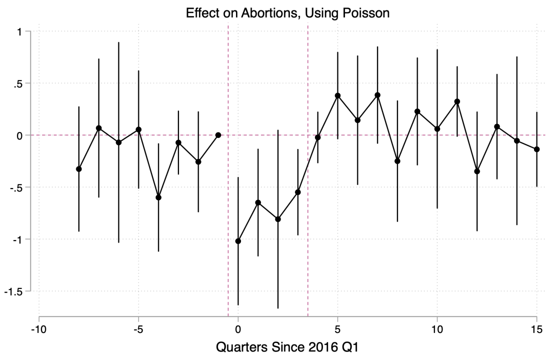
Figure 3
Estimated Effects on Births and Abortions, Poisson

(a) Births

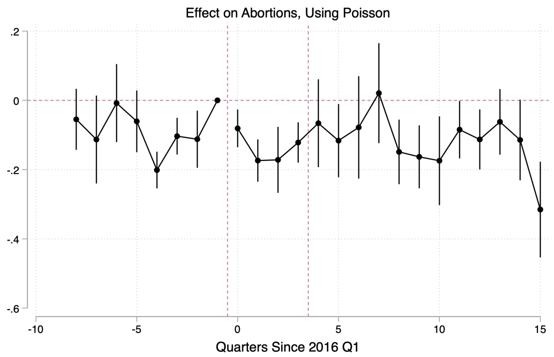
(b) Births, Shortage Counties



(c) Abortions



(d) Abortions, Shortage Counties



Notes: These figures plot the estimated effects of pharmacist prescription fo birth control on births and abortions using a Poisson model. The model controls for county and time fixed effects and population shares and unemployment rate. The right-hand panels restrict the treated group to *only* thoes that experienced a physician shortage.

Appendix

Table A1
Treatment Definitions

	Treated	Comparison
Zip Codes	389	801
Zip Codes with Provider Shortages	236	801
Counties	31	5
Counties with Provider Shortages	5	5

Zip Codes by County Treatment Status

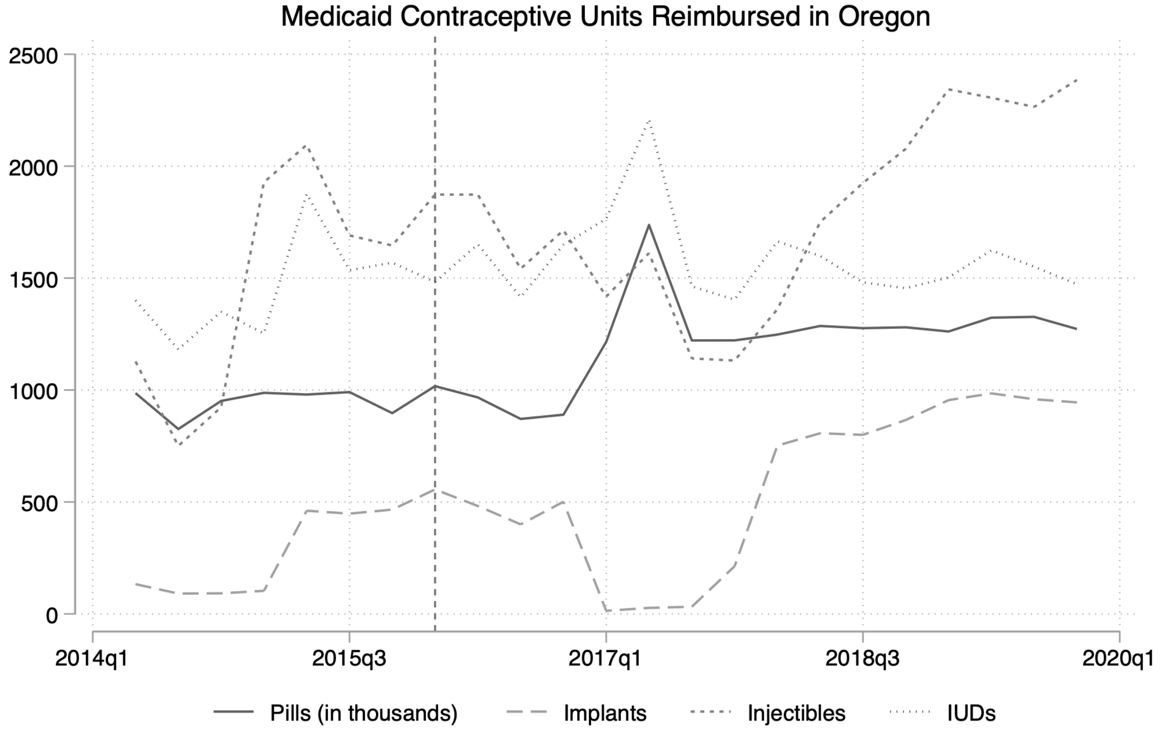
	Treated	Comparison
Untreated Zip Codes	764	37
Treated Zip Codes	380	9

Notes: This table shows the number of treated and comparison counties and zip codes. Since many zip codes are untreated but exist in counties that are treated, I argue that the zip code level analysis is preferred to the county-level.

Figure A1
Nationwide Birth Control Pharmacies

Notes: This image was taken from birthcontrolpharmacies.org on 14 Nov 2023. Prospective patients an zoom in on an area or enter their zip code to find a nearby pharmacy. Some entries are duplicates, inflating the numbers shown on the map; some pharmacists may not be listed but still be providing care, deflating the numbers shown on the map.

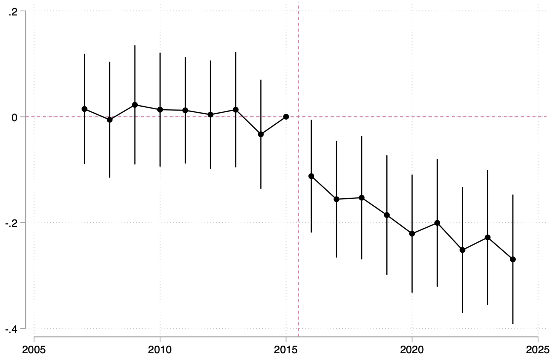
Figure A2
Medicaid Contraceptive Prescriptions in Oregon



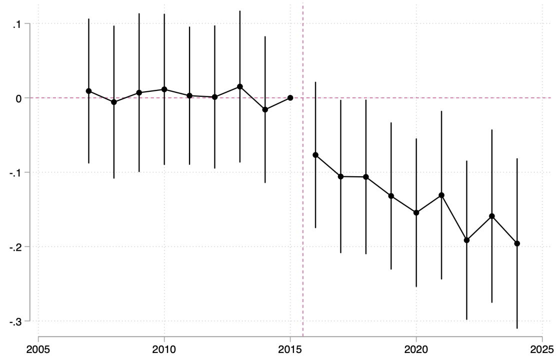
Notes: This figure plots state-level quarterly units of contraceptive medicine reimbursed by Medicaid in Oregon. The vertical line indicates the start of pharmacist prescription of birth control in Oregon

Figure A3
Testing Sensitivity of 'Provider Shortage' on Zip Code Births

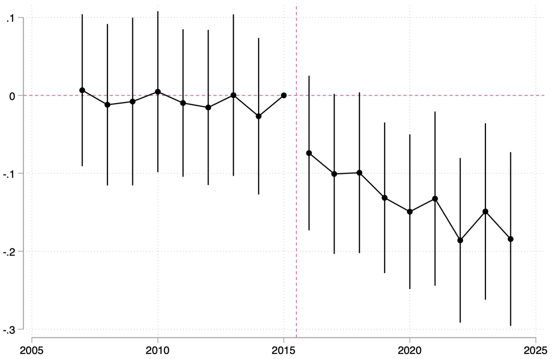
(a) Provider:Patient Ratio >= 603.39



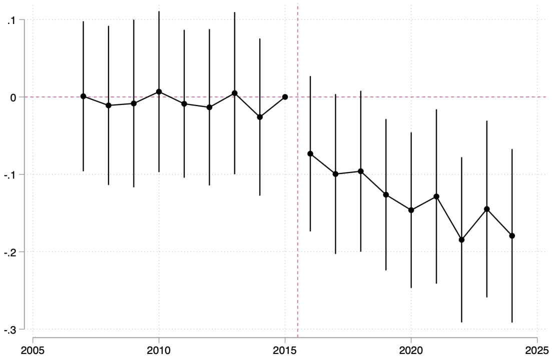
(b) Provider:Patient Ratio >= 445.12



(c) Provider:Patient Ratio >= 342.47



(d) Provider:Patient Ratio >= 238.61



Notes: These figures plot the estimated effects of pharmacist prescription for birth control on annual zip code births using a Poisson model. The model controls for county and time fixed effects and population shares and unemployment rate. Each figure uses a slightly more relaxed measurement of "provider shortage", with the provider to patient ratios identified as 'shortages' shown in the figure captions. The first graph, using the strictest definition, is my prefered specification.